Numerical Prediction of Particle Motion Trajectory under the Action of Acoustic Radiation Force

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ABSTRACT

Acoustic radiation force (ARF) for the non-destructive and non-contact manipulation of particles has been attracting great interest in material science, medical applications and so on. The numerical prediction of the motion trajectory of particles under the action of ARF is very helpful for the design, analysis, and physical understanding of acoustic manipulation devices. In this work, the liquid cylindrical particle is taken as the research object and that scattering coefficients are calculated theoretically to obtain its ARF. The motion trajectories of the particle are calculated. In order to make accurate predictions of the trajectory, we consider factors including the plane wave's incident angle, fluid viscosity, non-dimensional frequency kR, and initial velocity. Results show that the particle trajectory varies when any parameters influencing ARFs are modified. By adjusting these parameters, the particle's motion can be effectively predicted and controlled.

Keywords: particle motion trajectory; viscous fluid; liquid cylinder; acoustic radiation force

1. INTRODUCTION

Non-contact particle manipulation plays a crucial role in material science, medical applications and so on [1-2]. The requirements like trapping [3-4], rotation [5], separation [6], etc. which are an essential technique in numerous application fields, are increasing. Compared with other forces suitable for particle manipulation, acoustic manipulation utilizing acoustic radiation force (ARF) is often considered to have the merits, such as biocompatibility [7], contactless and non-invasive [8], spanning a larger scale of particle in length [9], and so on. ARF is a nonlinear effect of the acoustic field, which is the result of momentum transfer between the acoustic field and objects. When sound waves act on a particle that is freely suspended in the acoustic field, the particle will move under the action of ARF. The reported literatures mostly focus on the magnitude and distribution of ARF [10-12].

However, the aim of this work is to investigate the motion trajectory of a particle. The liquid cylinder suspended freely in a viscous fluid with a boundary is considered. Additionally, the scattering coefficients of the cylinder are calculated theoretically to obtain its ARF. Furthermore, the numerical predictions of the motion trajectories of the particle under the effects of the plane wave's incident angle, fluid viscosity, non-dimensional frequency kR, and initial velocity are carried out.

2. THEORY

2.1 Acoustic Scattering From a Free Fluid Cylinder

Consider that a fluid cylinder (radius R) is placed in viscous fluids with boundaries, with the separation distance denoted by d. A plane wave propagates in the θ incident direction. The range of θ is $-\pi/2 \le \theta \le \pi/2$. The external overall velocity potentials of the free cylinder are represented as [13]

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Fourth International Conference on New Materials, Machinery, and Vehicle Engineering (NMMVE 2025),
edited by Jinyang Xu, J. Paulo Davim, Proc. of SPIE Vol. 13953, 1395334

$$\Phi=\sum_{n=-\infty}^{\infty}\left[2A\cos\left(\alpha d\cos\theta+n\theta-\frac{n\pi}{2}\right)J_n\left(\alpha r_1\right)\right.$$
$$\left.+A_n^{(1)}H_n^{(1)}\left(\alpha r_l\right)+S_nJ_n\left(\alpha r_1\right)\right]e^{in\varphi_1}e^{-i\omega t} \tag{1}$$

$$\Psi=\sum_{n=-\infty}^{\infty}\left[B_n^{(1)}H_n^{(1)}\left(\beta r_l\right)+Q_nJ_n\left(\beta r_1\right)\right]e^{in\varphi_1}e^{-i\omega t}, \tag{2}$$

where

$$S_n=\sum_{m=-\infty}^{\infty}\left(-1\right)^m A_m^{(1)}H_{n-m}^{(1)}\left(2\alpha d\right),$$

$$Q_n=\sum_{m=-\infty}^{\infty}\left(-1\right)^m B_m^{(1)}H_{n-m}^{(1)}\left(2\beta d\right).$$

where Φ denotes the potential of longitudinal wave, while Ψ denotes that of transverse wave. The amplitude is defined as

$$A=\sqrt{2I_0/\rho_0c_0k^2}$$

where I_0 represents the incident wave's intensity, while ρ_0, c_0 stand for the density and sound velocity of liquids, respectively. Here, $k=\mathrm{Re}\left(\alpha\right)$. The longitudinal wave's wavenumber is given by

$$\alpha=\frac{\omega}{c_0}\left[1-i\omega\left(\lambda'+2\mu'\right)/\rho_0c_0^2\right]^{-1/2}$$

The function $J_n(\cdot)$ is the Bessel function of the first kind of order n, while $H_n^{(1)}(\cdot)=J_n(\cdot)+iY_n(\cdot)$ denotes the Hankel function of the first kind of order n. The shear wave's wavenumber is indicated by

$$\beta=\left(1+i\right)/\delta$$

Here, $\delta=\sqrt{2\mu'/\rho_0\omega}$ defines the penetration distance of the viscous wave. Parameters $\lambda'=\eta'-2\mu'/3$, η' and μ' correspond to the second viscosity coefficient and dynamic viscosity coefficient, respectively. $A_n^{(1)}$ and $B_n^{(1)}$, as scattering coefficients, are specified by conditions at the cylinder's lateral boundary.

For the case of an infinitely long, ideal liquid cylinder, when the sound wave is incident on it, only the longitudinal wave exists in the particle. The potential of longitudinal wave is written as

$$\overline{\Phi}_l^f=\sum_{n=-\infty}^{\infty}\overline{A}_nJ_n\left(\overline{k}_l^f r_1\right)e^{in\varphi_1}e^{-i\omega t}, \tag{3}$$

where $\overline{k}_l^f=\omega/c_l^f$ is the longitudinal and transverse wave number inside the cylinder and c_l^f is the longitudinal wave speed.

The boundary conditions at the surface of the fluid cylinder are expressed as

$$\begin{aligned} v_r\big|_{r=R}&=\overline{v}_r^f\big|_{r=R}, \\ \sigma_{rr}\big|_{r=R}&=\overline{p}\big|_{r=R}, \\ \sigma_{r\varphi}\big|_{r=R}&=0, \end{aligned} \tag{4}$$

where

$$v_r = \frac{\partial \Phi}{\partial r} + \frac{1}{r}\frac{\partial \Psi}{\partial \varphi},$$
$$\sigma_{rr} = -p_1 + 2\mu' \frac{\partial v_r}{\partial r} + \lambda' \left(\frac{\partial v_r}{\partial r} + \frac{1}{r}\frac{\partial v_\varphi}{\partial \varphi} + \frac{v_r}{r} \right),$$
$$\sigma_{r\varphi} = \mu' \left(\frac{1}{r}\frac{\partial v_r}{\partial \varphi} + \frac{\partial v_\varphi}{\partial r} - \frac{v_\varphi}{r} \right),$$
$$\overline{p} = i\omega\rho_0 \overline{\Phi}_l^f,$$
$$\overline{v}_r^f = \partial \overline{\Phi}_l^f / \partial r.$$

In which

$$p_1 = \rho_0 \left(\frac{\lambda' + 2\mu'}{\rho_0} \Delta - \frac{\partial}{\partial t} \right) \Phi + \frac{\rho_0}{2c_0^2} \left(\frac{\partial \Phi}{\partial t} \right)^2 - \frac{1}{2}\rho_0 \left(\overline{\nabla \Phi} \right)^2 - \frac{\lambda' + 2\mu'}{c_0^2} \frac{\partial \Phi}{\partial t} \Delta \Phi.$$

Substituting (1)-(3) into (4), after some arithmetic manipulations and sorting, we obtain a linear system of equations for the fluid cylinder

$$\left(\begin{bmatrix} C_r^f & D_r^f \\ C_\varphi^f & D_\varphi^f \end{bmatrix} - \begin{bmatrix} S_r^f & Q_r^f \\ S_\varphi^f & Q_\varphi^f \end{bmatrix} \begin{bmatrix} W_{1n}^f & O \\ O & W_{2n}^f \end{bmatrix} \right) \begin{bmatrix} A_n \\ B_n \end{bmatrix} = \begin{bmatrix} I_r^f \\ I_\varphi^f \end{bmatrix}. \tag{5}$$

where

$$A_n = \begin{bmatrix} A_{-n}^{(1)} & A_{-n+1}^{(1)} & \cdots & A_0^{(1)} & \cdots & A_{n-1}^{(1)} & A_n^{(1)} \end{bmatrix}^T,$$
$$B_n = \begin{bmatrix} B_{-n}^{(1)} & B_{-n+1}^{(1)} & \cdots & B_0^{(1)} & \cdots & B_{n-1}^{(1)} & B_n^{(1)} \end{bmatrix}^T,$$
$$W_{1n}^f = W_{1n},$$
$$W_{2n}^f = W_{2n},$$
$$I_r^f = \begin{bmatrix} I_{r,-n}^f & I_{r,-n+1}^f & \cdots & I_{r,0}^f & \cdots & I_{r,n-1}^f & I_{r,n}^f \end{bmatrix}^T,$$
$$I_\varphi^f = \begin{bmatrix} I_{\varphi,-n}^f & I_{\varphi,-n+1}^f & \cdots & I_{\varphi,0}^f & \cdots & I_{\varphi,n-1}^f & I_{\varphi,n}^f \end{bmatrix}^T,$$

in which

$$I_{r,n}^f = -2A\cos\left(\alpha d \cos\theta + n\theta - \frac{n\pi}{2} \right) \times \left(\frac{i\omega\rho_0 R^2}{2\mu'} \beta_{r,n} \overline{\gamma}_{r,n} - \varepsilon_{r,n} \overline{\beta}_{r,n} \right), \quad I_{\varphi,n}^f = -2iA\cos\left(\alpha d \cos\theta + n\theta - \frac{n\pi}{2} \right) \varepsilon_{\varphi,n}.$$

In (5), submatrics C_r^f, C_φ^f, D_r^f, D_φ^f, S_r^f, S_φ^f, Q_r^f and Q_φ^f are diagonal, with their respective diagonal elements given by $C_{r,n}^f$, $C_{\varphi,n}^f$, $D_{r,n}^f$, $D_{\varphi,n}^f$, $S_{r,n}^f$, $S_{\varphi,n}^f$, $Q_{r,n}^f$ and $Q_{\varphi,n}^f$. The matrix O is a zero matrix with dimensions identical to those of W_{1n}^f and W_{2n}^f. These diagonal matrices have elements given by:

$$C^f_{r,n} = \frac{i\omega\rho_0 R^2}{2\mu'}\gamma_{r,n}\overline{\gamma}_{r,n} - a_{r,n}\overline{\beta}_{r,n},$$

$$C^f_{\varphi,n} = i\frac{i\omega\rho_0 R^2}{2\mu'}a_{\varphi,n},$$

$$D^f_{r,n} = \frac{-\omega\rho_0 R^2}{2\mu'}\delta_{r,n}\overline{\gamma}_{r,n} - ib_{r,n}\overline{\beta}_{r,n},$$

$$D^f_{\varphi,n} = b_{\varphi,n},$$

$$S^f_{r,n} = \frac{-i\omega\rho_0 R^2}{2\mu'}\beta_{r,n}\overline{\gamma}_{r,n} + d_{r,n}\overline{\beta}_{r,n},$$

$$S^f_{\varphi,n} = -id_{\varphi,n},$$

$$Q^f_{r,n} = \frac{\omega\rho_0 R^2}{2\mu'}\chi_{r,n}\overline{\gamma}_{r,n} + ie_{r,n}\overline{\beta}_{r,n},$$

$$Q^f_{\varphi,n} = -e_{\varphi,n},$$

$$\overline{\gamma}_{r,n} = J_n\left(\overline{k}_l^f R\right),$$

$$\overline{\beta}_{r,n} = nJ_n\left(\overline{k}_l^f R\right) - \overline{k}_l^f RJ_{n+1}\left(\overline{k}_l^f R\right),$$

$$\gamma_{r,n} = nH_n^{(1)}(\alpha R) - \alpha R_1 H_{n+1}^{(1)}(\alpha R),$$

$$a_{r,n} = \left(n^2 - n - 0.5\beta^2 R^2\right)H_n^{(1)}(\alpha R) + \alpha RH_{n+1}^{(1)}(\alpha R),\ a_{\varphi,n} = \left(n^2 - n\right)H_n^{(1)}(\alpha R) - n\alpha RH_{n+1}^{(1)}(\alpha R),$$

$$\beta_{r,n} = nJ_n(\alpha R) - \alpha RJ_{n+1}(\alpha R),$$

$$\delta_{r,n} = nH_n^{(1)}(\beta R),$$

$$b_{r,n} = \left(n^2 - n\right)H_n^{(1)}(\beta R) - n\beta RH_{n+1}^{(1)}(\beta R),$$

$$b_{\varphi,n} = \left(n^2 - n + 0.5\beta^2 R^2\right)H_n^{(1)}(\beta R) - \beta RH_{n+1}^{(1)}(\beta R),\ d_{r,n} = \left(n^2 - n - 0.5\beta^2 R^2\right)J_n(\alpha R) + \alpha RJ_{n+1}(\alpha R),$$

$$d_{\varphi,n} = \left(n^2 - n\right)J_n(\alpha R) - n\alpha RJ_{n+1}(\alpha R),$$

$$\chi_{r,n} = nJ_n(\beta R),$$

$$e_{r,n} = \left(n^2 - n\right)J_n(\beta R) - n\beta RJ_{n+1}(\beta R),$$

$$e_{\varphi,n} = \left(n^2 - n + 0.5\beta^2 R^2\right)J_n(\beta R) - \beta RJ_{n+1}(\beta R),$$

The unknown scattering coefficients for the fluid cylinder are computed by solving (5).

2.2 Motion of the freely moving cylinder

The prediction of trajectories of free objects is very important in particle manipulation. The dynamics of the free cylinder subjected to ARFs can be formulated using Newton's second law as

$$m\ddot{\mathbf{r}} = \mathbf{F} + \mathbf{F}^{drag} \tag{6}$$

where $\ddot{\mathbf{r}} = d^2\mathbf{r}/dt^2$ characterizes the cylinder's acceleration, $m = \pi R^2 \rho_1$ is the mass of a unit length of the cylinder, $\mathbf{F}$ is the ARFs calculated by (23) and (24) in [13], but the scattering coefficients in the formula is replaced by that obtained from (5), and $\mathbf{F}^{drag} = -1/2\, \rho_0 \dot{\mathbf{r}}^2 R C_D$ is the viscous drag force of a two-dimensional cylinder [14, 15], with $\dot{\mathbf{r}} = d\mathbf{r}/dt$ the cylinder velocity, the position vector is given by $\mathbf{r} = x\mathbf{e}_\mathbf{x} + y\mathbf{e}_\mathbf{y}$. In the subsequent simulation, the C_D, as clearly stated in [14], is given as (3.4) in [14] due to the small Reynolds number $\mathrm{Re} = R\dot{\mathbf{r}}/\nu$, namely

$$C_D = \frac{4\pi\zeta}{\mathrm{Re}}\left(\frac{2}{1+\sqrt{1+4.3\zeta^2}}\right)$$

and $\zeta = \left[0.5 - \gamma - \log\left(\mathrm{Re}/4\right)\right]^{-1}$, with $\gamma = 0.577216$.

The trajectories of the free particle in varying conditions are obtained by (6) combining Runge-Kutta methods.

3. NUMERICAL RESULTS AND DISCUSSIONS

In this section, the numerical predictions of a fluid cylinder trajectories under the action of ARF under different conditions are illustrated intuitively. To ensure convergence to various situations, $N_{\max} = M_{\max} = 30$ are chosen. The viscous liquid parameters and the physical parameters of the free cylinder used in the calculations are listed in Table I and Table II respectively.

TABLE I. The parameters of the viscous fluid in the calculations [16].

Material	***Density ρ_0* (kg/m^3)**	***Sound speed c_0* (m/s)**	***Dynamic viscosity μ'* (Pa·s)**
Water	1000	1500	0.001
Glycerin	1260	1900	1.48

TABLE II. The physical parameters of the free cylinder used in the calculations [4].

Material	***Density ρ_0* (kg/m^3)**	***Compressional velocity c_l* (m/s)**
Oleic acid (OA)	938	1450

The results are shown in Figs. 1-4. Arrows mark the direction of particle motion. Fig. 1 displays the trajectories of the OA cylinder for varied incident angles. The $R = 0.01\mathrm{mm}$ and $f = 2\mathrm{MHz}$. The solid, dash-dot and dashed lines correspond to different θ. As seen from the Fig. 1, the trajectories of the particles depend on the incident angle θ. The particle's behavior can be controlled by varying the incident angle. Furthermore, the trajectories of the OA cylinder for varied kR are presented in Fig. 2. Even when all external conditions are held constant, the particle trajectories vary with kR, suggesting that ARF can be utilized to selectively separate and filter particles. In addition, initial velocity v_0 and liquid viscosity also play critical roles in determining the trajectories. The results are given in Figs. 3–4. These Figs. reveal that trajectories vary when any parameters influencing ARFs are modified. So, the particle trajectory subjected to ARF can be predicted and regulated according to the known conditions. This enables both the individual particle capture and their assembly into precisely clusters for future applications.

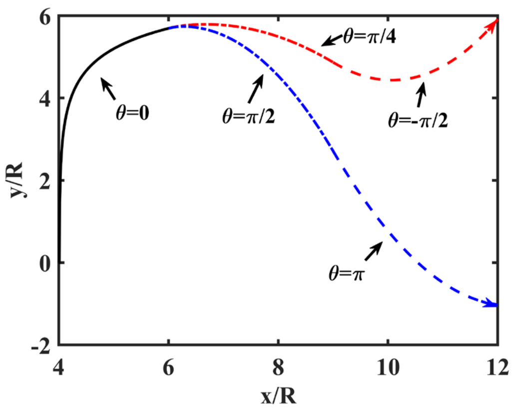

Figure 1. Trajectories of the OA cylinder for varied incident angles with $R = 0.01\text{mm}$ and $f = 2\text{MHz}$

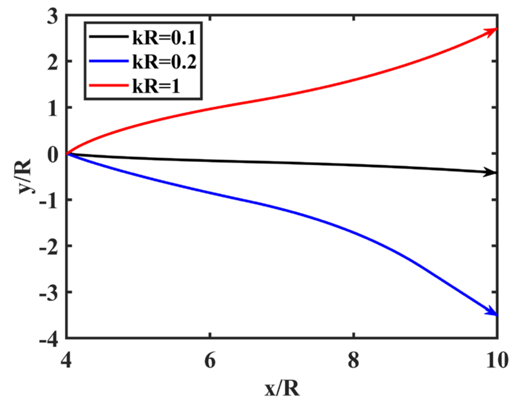

Figure 2. Trajectories of the OA cylinder for varied kR .

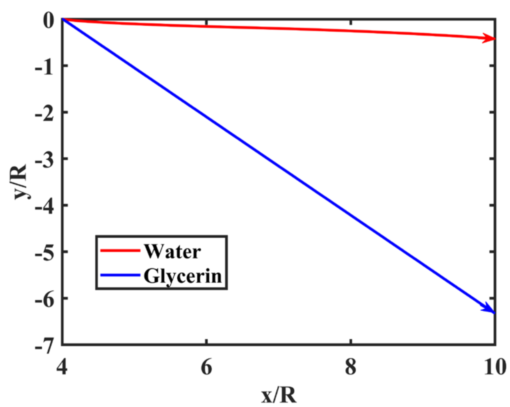

Figure 3. Trajectories of the OA cylinder in two different viscous liquids (glycerin and water) for $\theta = \pi/4$ and, $f = 2\text{MHz}$.

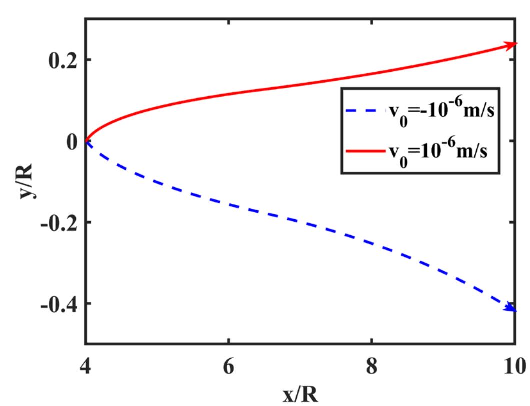

Figure 4. Trajectories of the OA cylinder for different initial velocity and $\theta = \pi/4$ with $R = 0.01mm$, $f = 2\text{MHz}$.

4. CONCLUSIONS

This work presents the scattering coefficients of a free fluid cylinder suspended in viscous fluids with boundaries subjected to plane wave incidence at an arbitrary angle theoretically to obtain its ARF. Following Newton's second law, trajectories of the free fluid cylinder under different conditions including the plane wave's incident angle, fluid viscosity, non-dimensional frequency kR, and initial velocity are calculated combining Runge-Kutta methods. Results show that the particle trajectory varies when any parameters influencing ARFs are modified. By adjusting these parameters, the particle's motion can be effectively predicted and controlled, enabling not only individual capture but also the formation of well-organized clusters for future applications. This work is very helpful for the design, analysis, and physical understanding of acoustic manipulation devices.

ACKNOWLEDGMENT

This work was supported by National Natural Science Foundation of China (Grant Nos. 12204119, 12404534, 12264008), Guizhou Provincial Science and Technology Foundation (No. ZK[2023]249), and the Natural Science and Technology Foundation of Guizhou Province (No. [2024]054).

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